

Response to reviewers

QAOA-Simulated Global Pressure-Based Energy Optimization for Traffic Signal Control

Author: Arnav Jha | Editor: Shireesh Apte

Dear Dr. Apte and Reviewers,

Thank you for the detailed evaluation of the manuscript. The revision expands the study to include a classical global comparison, exact energy benchmarking, a nine-intersection network, additional adaptive traffic controllers, repeated-run statistical analyses, and QAOA sensitivity experiments. The responses below identify the corresponding manuscript sections and explain the scope and limitations of the evidence.

Each open-response comment is reproduced verbatim and answered separately, including the introductory and concluding comments. Page references refer to the supplied 86-page revised PDF. Items marked “Correction required before resubmission” identify changes still needed before the response can represent a completed revision; they should be resolved and the associated response updated before submission. Figure numbers identify the manuscript’s intended figures, several of which remain placeholders in the supplied PDF.

Reviewer 1

Rating scale questions

Verbatim rating statement The paper makes a significant contribution to scholarship.

Verbatim rating statement The literature review was thorough given the objectives and content of the article.

The supplied scale reads “Strongly Disagree”, “Disagree”, “Neutral”, “Agree”, and “Strongly Agree”. Checkmarks were supplied, but their column alignment was not retained in the pasted feedback, so no rating category is inferred. The substantive concerns underlying these statements are addressed below.

R1.1 Scientific contribution and overall experimental rigor

Reviewer comment verbatim The manuscript presents an interesting integration of pressure-based traffic optimization with the Quantum Approximate Optimization Algorithm (QAOA), and the pressure-based energy formulation constitutes the principal scientific contribution of the work. The mathematical mapping from traffic pressure to an Ising Hamiltonian is elegant and internally consistent. However, the current experimental design does not yet provide sufficient evidence to conclude that the reported improvements arise specifically from the use of QAOA rather than from more general properties of global optimization. Several additional analyses and controls would substantially strengthen the manuscript and improve the scientific rigor of the conclusions.

Response The reviewer’s distinction between the pressure-based objective and the method used to search it is accepted. The revision separates Pressure-Based Energy Optimization from QAOA Global Optimization and evaluates both against broader traffic-control benchmarks. Relative to the original submission, the manuscript adds a classical global comparison, exact optimum diagnostics, 3×3 experiments, energy–traffic correlations, bootstrap intervals, effect sizes, and parameter and synthetic-noise sensitivity analyses.

The revised interpretation is conditional: QAOA can produce different closed-loop traffic trajectories, but it does not improve on exhaustive enumeration as an optimizer of the same instantaneous Hamiltonian. Neither computational quantum advantage nor a benefit unique to quantum sampling has been established.

Location in revised manuscript Introduction, pp. 7–9; Sections 2.9–2.10, pp. 36–45; Sections 3.4–3.9, pp. 61–71; Section 4.2, pp. 77–80; Section 5, pp. 80–83.

Correction required before resubmission Resolve the classical-controller implementation mismatch described in R1.2 and replace remaining statements that the comparison “isolates the quantum contribution” with the narrower claim that it compares the implemented global search procedures. A classical stochastic sampling control would be needed to test whether the observed behavior is specific to QAOA.

R1.2 Separate optimization scope from optimization method

Reviewer comment verbatim The most significant methodological concern is that the current comparison simultaneously changes two experimental variables. The classical controller performs local optimization, whereas the QAOA controller performs global optimization over the entire network-wide Ising objective. Consequently, the observed performance improvements cannot be uniquely attributed to the quantum optimization algorithm because optimization scope (local versus global) and optimization strategy (classical versus QAOA) change simultaneously. The present design therefore contains an important confounding variable. A scientifically stronger comparison would include a classical global optimizer that minimizes exactly the same network-wide Ising Hamiltonian as the QAOA controller. Appropriate benchmarks include exhaustive enumeration (for the current network), simulated annealing, tabu search, branch-and-bound, mixed-integer optimization, or another established global optimization algorithm. Such comparisons would isolate the incremental contribution of the QAOA search strategy itself rather than demonstrating only that global optimization outperforms local optimization.

Response A Classical Global controller has been added alongside Classical Local and QAOA Global. Section 2.9 defines the common objective, including pressure bias, switching strength $\lambda = 10$, and neighbor coupling $J = 2$. This provides a global classical comparison without changing the spatial scope of the optimization. Tables 14–17 report the traffic results, and Table 21 reports QAOA-minus-Classical-Global differences and uncertainty.

The current code, however, requires a correction to the manuscript’s description of this benchmark. In both `grid_2x2/2_classical_global/a7.py` and `grid_3x3/2_classical_global/a7.py`, the phase recommendation comes from `quantum_decision` in `annealer_classical.py`. That routine performs stochastic simulated annealing with eight restarts and 512 iterations per restart. Exhaustive enumeration is then used to calculate an energy diagnostic, rather than to choose the applied recommendation. Thus, the workspace supports a matched-objective classical simulated-annealing comparison, but does not substantiate calling the deployed controller exact and deterministic.

Location in revised manuscript Section 2.1.4 and Table 1, p. 12; Sections 2.9.2–2.9.3, pp. 37–38; Section 2.9.8 and Table 11, p. 42; Tables 14–17, pp. 46, 49, 55 and 58; Section 3.6 and Table 21, pp. 67–68.

Correction required before resubmission Either document the exact code version that generated the reported exact-controller results, or run a controller that actually applies an enumerated minimizer with a stated tie-breaking rule and regenerate the affected results. If simulated annealing is retained, label it accurately and quantify its stochastic variability. Identical input definitions and demand do not mean that independent closed-loop controllers encounter identical evolving traffic states; reserve same-state comparisons for the decision-level energy benchmark.

R1.3 Exhaustive energy benchmark on the four-intersection network

Reviewer comment verbatim This issue is particularly important because the current experiments are performed on a 2x2 traffic network comprising only four intersections, corresponding to only 16 possible global signal configurations. For a problem of this size, the global optimum can be obtained exactly through exhaustive classical enumeration. Consequently, the manuscript should report the exact minimum Ising energy

obtained by exhaustive search and compare it directly with both the classical heuristic and the QAOA solution. This would establish (i) whether QAOA actually converges to the global optimum, (ii) the probability that QAOA recovers the true optimum across repeated stochastic runs, and (iii) the optimality gap when it does not. Without this benchmark it is difficult to assess the actual optimization performance of the proposed quantum algorithm. Furthermore, because exhaustive optimization is computationally trivial for this network, it represents the appropriate gold standard rather than merely another optional comparison.

Response The revision adds exhaustive enumeration of all 16 configurations for each evaluated four-intersection QAOA decision and all 512 configurations for the nine-intersection analysis. The diagnostic compares the QAOA recommendation with the minimum of the same instantaneous energy function, using $\Delta E = E_{\text{selected}} - E_{\text{min}}$. It is separate from the deployed classical controller discussed in R1.2.

At $\alpha = 0.7$, Table 18 reports mean exact and QAOA energies of -92.131 and -92.057 on the 2×2 grid, with a mean gap of 0.074 and recovery of 99.40% (bootstrap 95% CI $99.30\text{--}99.50\%$). For 3×3 , the corresponding values are -206.844 , -183.520 , a gap of 23.323 , and recovery of 41.42% ($39.85\text{--}43.09\%$). These are averages of run-level decision averages across changing Hamiltonians, rather than one fixed problem's minimum. Recovery means selecting any energy-minimizing configuration, including tied minima; it is not a proof of convergence as circuit depth increases.

Location in revised manuscript Sections 2.9.3 and 2.9.7, pp. 37–38 and 41; Section 3.4 and Table 18, pp. 61–63; intended Figures 17–19.

Correction required before resubmission Insert Figures 17–19. The current Table 18 directly compares QAOA with exact energy but does not supply a same-state Classical Local heuristic energy series. Add that diagnostic to complete the reviewer's three-way energy comparison, and disclose the numerical optimum tolerance used in code ($1e-9$ for the recovery test) rather than implying exact floating-point equality.

R1.4 Larger network evaluation

Reviewer comment verbatim The manuscript would also benefit from evaluating the methodology on a larger traffic network. While a 5×5 grid would certainly be informative, I believe that extending the study to at least a 3×3 grid would provide the most appropriate balance between computational feasibility and scientific value. A 3×3 network already increases the optimization problem substantially while remaining sufficiently tractable for comparison against strong classical optimization methods. Demonstrating that the proposed framework continues to provide benefits beyond the smallest non-trivial network would considerably strengthen the manuscript's claims regarding practical applicability. If evaluation on a larger grid is beyond the intended scope of the current work, this limitation should at least be discussed explicitly.

Response The evaluation now includes a 3×3 grid with nine binary signal variables and 512 global configurations. Both the 600-second saturated α sweep and the 24-hour Seattle-derived demand experiment are evaluated on this larger network, with the same nine controller categories used for 2×2 .

At $\alpha = 0.7$, Table 15 reports throughput of 980 vehicles for QAOA, 944 for Classical Local, 940 for the controller labeled Classical Global, and 925 for Max-Pressure. QAOA waiting time is 43.75 s versus 47.16 s for Max-Pressure. The larger-grid results also expose limitations: exact-optimum recovery falls sharply, and some throughput gains accompany higher completed-vehicle delay. Section 5 explicitly limits generalization to small regular grids and distinguishes Seattle-derived demand from the actual Seattle road network.

Location in revised manuscript Sections 2.2–2.2.2, pp. 12–13; Sections 2.7.9–2.7.10, pp. 31–32; Table 15, p. 49; Sections 3.2–3.3, pp. 53–60; Table 17, p. 58; Section 5, pp. 81–83.

Correction required before resubmission Retain the benchmark qualification in R1.2 and insert the intended scaling and Seattle figures. Describe 3×3 as a controlled extension, not evidence of city-scale feasibility.

R1.5 Synthetic quantum noise

Reviewer comment verbatim Another important concern is the evaluation of QAOA under an entirely idealized quantum computational model. The current simulations assume perfect quantum operations and do not consider gate infidelity, measurement errors, decoherence, depolarizing noise, parameter uncertainty, circuit connectivity constraints, transpilation overhead, or other realistic characteristics of present-day quantum hardware. The issue is not that the algorithm is simulated rather than executed on physical hardware—simulation is entirely appropriate for methodological development—but rather that the manuscript provides no assessment of how sensitive the reported performance gains are to deviations from ideal quantum execution. The conclusions would therefore be strengthened considerably by a systematic sensitivity analysis in which synthetic quantum noise is introduced into the simulation. Examples include gate perturbations, measurement noise, depolarizing channels, phase-flip or bit-flip errors, or parameter uncertainty in the QAOA rotation angles. Such analyses would determine whether the reported improvements remain robust under realistic execution conditions or whether they depend critically upon idealized assumptions.

Response The revision adds three synthetic perturbation families on the 3×3 saturated network at $\alpha = 0.7$: Gaussian angle uncertainty with standard deviations 0.025, 0.050 and 0.100 rad; depolarizing probabilities 0.5%, 1% and 2%; and symmetric readout bit-flip probabilities 1%, 2% and 5%. Table 23 reports traffic metrics, optimality gaps and recovery for 10 runs per noisy condition and a 20-run ideal reference. Its intervals are Student-t intervals, not bootstrap intervals.

The implementation in `grid_3x3/3_quantum/annealer_noise.py` perturbs γ and β , attaches depolarizing channels to modeled one- and two-qubit gates, and applies readout flips to sampled bits. Reported travel times span approximately 195.05–200.35 s under angle uncertainty, compared with 197.50 s for the ideal reference. Optimization quality changes more markedly and is not monotonic with noise. These findings support limited sensitivity statements for the tested perturbations; they do not establish hardware fault tolerance, equivalence of noisy and ideal performance, or persistence of every gain over classical baselines.

Location in revised manuscript Section 2.10.4, p. 45; Sections 3.8–3.9 and Table 23, pp. 70–71; Section 4.2.2, pp. 79–80; Section 5, pp. 82–83; intended Figures 25–26.

Correction required before resubmission Insert the noise figures and merge the duplicate Sections 3.8 and 3.9. Expand the methods to state the perturbation placement, calibration procedure and seed handling. Retain explicit exclusions for device-calibrated decoherence, connectivity, latency and hardware overhead; overlapping intervals alone do not establish robustness by an equivalence test.

R1.6 Hyperparameter and measurement sensitivity

Reviewer comment verbatim Related to this, the manuscript currently fixes the QAOA hyperparameters without demonstrating the robustness of the reported results. Performance should ideally be evaluated over a range of QAOA parameters (e.g., rotation angles, circuit depth, and measurement shots) to demonstrate that the observed improvements do not depend upon a particular parameter selection. Similarly, sensitivity analyses varying the number of measurement shots would provide insight into the stability of the optimization procedure.

Response The revision describes a 200-candidate calibration grid over γ , β and p , evaluated with 512 shots, and adds a separate one-at-a-time sensitivity experiment on 3×3 at $\alpha = 0.7$. Table 22 varies γ over $4\pi/9$, $7\pi/9$ and π ; β over $\pi/9$, $\pi/6$ and $5\pi/18$; depth over 1, 2 and 3; and shots over 200, 512 and 1,000. The reference is $\gamma = \pi$, $\beta = \pi/6$, $p = 2$ and 512 shots, with 10 independent runs per setting.

Increasing depth from 1 to 3 raises exact recovery from 27.23% to 44.35% and decreases mean gap from 31.457 to 16.663. Increasing shots from 200 to 1,000 raises recovery from 27.63% to 48.13% and decreases gap from 32.319 to 19.231. Traffic outcomes are less sensitive and non-monotonic. The revision therefore separates improved approximation quality from improved traffic performance and acknowledges that the one-at-a-time design does not test parameter interactions.

Location in revised manuscript Sections 2.9.4–2.9.6, pp. 38–41; Section 3.7 and Table 22, pp. 68–69; Section 5, pp. 82–83; intended Figures 23–24.

Correction required before resubmission Insert Figures 23–24 and specify the actual calibration frequency for each experiment rather than referring only to a schedule defined by the simulation mode. Table 22 uses Student-t intervals; avoid describing all sensitivity intervals as bootstrap estimates.

R1.7 Relationship between Ising energy and traffic outcomes

Reviewer comment verbatim A further issue concerns the relationship between the Ising objective function and actual traffic performance. The manuscript implicitly assumes that minimizing the pressure-based Ising Hamiltonian necessarily translates into improved traffic metrics such as waiting time, travel time, or throughput. However, no analysis is presented demonstrating that lower Ising energies consistently correspond to improved network performance. It would therefore be valuable to quantify the relationship between optimized Ising energy and downstream traffic metrics through correlation or regression analyses. Such an analysis would validate the proposed objective function itself rather than merely the optimization algorithm applied to it.

Response Section 3.5 adds a within-condition correlation analysis of run-average optimality gap against travel time, waiting time and throughput. Variables are centered within each α condition to reduce confounding from different turning regimes. Each grid contributes 220 run records: 20 runs across 11 α values.

Table 19 reports 2×2 correlations of 0.29 for travel time, 0.28 for waiting time and −0.28 for throughput (all $p < 0.001$). On 3×3, travel time and waiting time show little association ($r = -0.05$, $p = 0.458$; $r = -0.03$, $p = 0.631$), while throughput has a weak negative correlation ($r = -0.15$, $p = 0.032$). The evidence therefore supports modest alignment on the smaller grid and a weaker relationship on the larger grid. It does not support the original implication that minimizing instantaneous energy necessarily minimizes future delay.

The analysis measures approximation gap, not the causal effect of changing absolute energy while holding traffic state fixed. The saved records summarize decisions over whole runs; these are run-level associations rather than time-lagged, decision-level predictions.

Location in revised manuscript Section 3.5 and Table 19, pp. 63–64; Sections 4.1.2 and 4.2, pp. 75–80; Section 5, pp. 81–83; intended Figure 20.

Correction required before resubmission Insert Figure 20. Describe the findings as correlations with optimality gap, and avoid causal language or universal objective validation. Clarify whether the exploratory correlation p-values are adjusted for multiple testing.

R1.8 Bootstrap intervals effect sizes and distributions

Reviewer comment verbatim The statistical analysis should also be strengthened. At present the manuscript relies primarily upon averages, standard deviations, and significance testing across repeated simulation runs. Given that QAOA is itself a stochastic optimization algorithm, I recommend reporting bootstrap confidence intervals for all principal outcome measures, including waiting time, travel time, throughput, and energy reduction. Confidence intervals on the improvement attributable to QAOA would provide substantially more information than p-values alone. In addition, reporting effect sizes (e.g., Cohen's d , Hedges' g , or appropriate

non-parametric alternatives) would allow readers to assess the practical significance of the observed improvements rather than relying exclusively on statistical significance. Distributional visualizations such as boxplots or violin plots would further illustrate the variability of the optimization outcomes.

Response Tables 14–15 and 20 add percentile-bootstrap 95% intervals for the principal saturated traffic outcomes. Table 21 adds native-unit QAOA-minus-Classical-Global differences, bootstrap intervals, and one-sample Hedges' g , including full-day Seattle outcomes. The table-generation script uses 10,000 run-level resamples; the standardized effect divides the mean difference by QAOA's run-level sample standard deviation with a small-sample correction. This is not a pooled two-group effect size.

For illustration, Table 21 reports a 3×3 $\alpha = 0.7$ throughput difference of +40 vehicles (95% CI 27–54; $g = 1.22$), while the travel-time difference is +1.03 s (–1.60 to 3.41) and waiting-time difference +0.21 s (–0.14 to 0.51). At $\alpha = 0.5$, the throughput difference is +122 vehicles but travel time increases by 13.31 s. The manuscript therefore reports tradeoffs rather than selecting only favorable outcomes. Intended Figures 18 and 21 show run-average gap and traffic-outcome distributions.

Location in revised manuscript Tables 14–15, pp. 46 and 49; Section 3.6, Tables 20–21, pp. 64–68; intended Figures 18, 21 and 22.

Correction required before resubmission The request is only partially complete: add bootstrap intervals for energy reduction, which is recorded in the code but not reported with bootstrap uncertainty in these tables. Resolve the Section 2.10.3 claim of 50 2×2 runs: the current JSON files and tables use 20 per α for both grids. Insert the distribution figures. Uncertainty relative to a deterministic classical benchmark is valid only once R1.2 is resolved; otherwise repeat and include the classical stochastic runs. These intervals condition on the fixed demand scenario and do not measure variation across independent traffic-demand realizations or RL training seeds.

R1.9 Probability of optimum recovery

Reviewer comment verbatim Because the exact global optimum is known for the current network, an especially informative statistical measure would be the probability that QAOA recovers the global optimum, together with bootstrap confidence intervals for this probability. Alternatively, the manuscript could report the distribution of the optimality gap between QAOA and the exact optimum across repeated runs. These metrics would directly evaluate the optimization algorithm itself rather than relying solely on downstream traffic metrics.

Response Both diagnostics have been added. Section 3.4.2 describes the distribution of run-average gaps, and Section 3.4.3 reports optimum recovery with bootstrap intervals. At $\alpha = 0.7$, recovery is 99.40% (99.30–99.50%) for 2×2 and 41.42% (39.85–43.09%) for 3×3 . The median run-average gaps are 0.067 and 24.078, respectively.

Each run's recovery fraction is calculated from its optimum-hit count divided by its number of evaluated decisions. Bootstrap resampling is across independent run records, preserving the within-run dependence of sequential decisions. These are decision-recovery fractions averaged across runs, not the probability that every decision in a simulation is optimal, nor the probability that an individual measurement shot returns an optimum.

Location in revised manuscript Section 2.9.7, p. 41; Sections 3.4.2–3.4.3 and Table 18, pp. 61–63; intended Figures 18–19.

Correction required before resubmission Insert the figures and retain the run-level unit of analysis in their captions. Do not describe the stored run-average gap distributions as distributions of individual decision gaps.

R1.10 Established adaptive traffic controllers

Reviewer comment verbatim Finally, the manuscript would benefit from broader benchmarking against established adaptive traffic-signal control methods. Although the current comparisons against fixed-time, queue-based control, and the proposed pressure-based controller are useful, additional comparisons with established adaptive controllers such as Max-Pressure, SCOOT, SCATS, PressLight, CoLight, MPLight, or other state-of-the-art adaptive traffic signal algorithms would place the proposed methodology within the context of the existing literature and provide a clearer indication of its practical significance.

Response The revised comparison includes Fixed-Time, Classical Local, Classical Global, QAOA Global, Max-Pressure, SCOOT-inspired control, PressLight, MPLight and CoLight. Sections 2.4–2.6 explain the added adaptive controllers and their implementation. The RL controllers are adaptations to the common two-phase action space, trained on 2×2 at $\alpha = 0.7$ and then evaluated with frozen policies and exploration disabled. The SCOOT implementation is explicitly described as inspired by published principles rather than a certified commercial deployment. SCATS was not evaluated.

The expanded results show that QAOA does not dominate every method or metric. For example, at 3×3 $\alpha = 1.0$, Table 15 reports Max-Pressure throughput of 1,096 versus 1,015 for QAOA. The discussion also recognizes strong delay results from CoLight and MPLight and the performance of cycle-based control in the daily scenario. These comparisons place the proposed formulation in a broader context while retaining implementation and training-transfer limitations.

Location in revised manuscript Sections 2.4–2.6, pp. 17–24; Section 2.10.3, pp. 44–45; Tables 14–17, pp. 46, 49, 55 and 58; Sections 4–5, pp. 73–83.

Correction required before resubmission Use “SCOOT-inspired” consistently in tables and claims, retain the adaptation and frozen-policy qualifications, and avoid implying a comprehensive evaluation of commercial or fully tuned state-of-the-art systems. Correct the abstract’s inconsistent count: the listed total is nine controllers.

R1.11 Overall assessment and requested strengthening

Reviewer comment verbatim Overall, I believe the proposed pressure-based Ising formulation is scientifically interesting and the manuscript has considerable potential. However, the current evidence does not yet isolate the contribution of the quantum optimization algorithm from the broader benefits of global optimization, nor does it quantify the robustness of the reported improvements to realistic quantum uncertainties. Incorporating an exact classical global optimization benchmark, extending the evaluation beyond the current 2×2 network, performing sensitivity analyses under synthetic quantum noise, strengthening the statistical analyses with bootstrap confidence intervals and effect sizes, evaluating the probability of recovering the true optimum, and validating the relationship between Ising energy and traffic performance would substantially increase the rigor, credibility, and long-term impact of the study.

Response The revision addresses the requested directions through the additions detailed in R1.2–R1.10: global classical comparison and exact energy diagnostics, the 3×3 evaluation, synthetic perturbation and hyperparameter tests, run-level statistical uncertainty and effects, optimum-recovery analysis, and energy-gap correlations. The discussion now separates the usefulness of the energy objective from the quality of its approximate optimization and states that the observed traffic advantages are conditional.

The central unresolved issue is the difference between an exact energy diagnostic and an exact controller. Resolving that implementation and provenance issue, completing the missing figures and energy-reduction uncertainty, and narrowing quantum-specific attribution are necessary before a complete response to this overall assessment can be claimed.

Location in revised manuscript Sections 2.9–2.10, pp. 36–45; Sections 3.2–3.9, pp. 53–71; Sections 4.2–5, pp. 77–83.

Correction required before resubmission Complete the outstanding items in R1.2, R1.3 and R1.8 before representing all requested controls and statistical analyses as finished.

Reviewer 2

Rating scale questions

Verbatim rating statement The paper makes a significant contribution to scholarship.

Verbatim rating statement The literature review was thorough given the objectives and content of the article.

The same five-point scale and checkmark-alignment limitation described for Reviewer 1 apply. The responses below address the reviewer’s six open-response comments.

R2.1 Reference links and citation accuracy

Reviewer comment verbatim Please verify that all links to the references point to the correct source.

Response The revised reference list includes 19 embedded external links, using DOI, publisher, repository or official-data destinations. The reference audit identifies several destinations that match their cited works, including the original SCOOT report, the AAAI “Toward a Thousand Lights” paper, the two arXiv algorithm papers, the United Nations urbanization report and the Frontiers Ising-formulation paper.

Full compliance cannot yet be claimed. Reference [1] links to the correct 2025 Urban Mobility Report but its author list is inconsistent with the linked report: the report lists David Schrank, Luke Albert, Kartikeya Jha and Bill Eisele, whereas the manuscript lists David Schrank, Bill Eisele and Tim Lomax. Reference [14] links to the project’s GitHub CSV rather than the original City of Seattle dataset. Several publisher links could not be fully inspected because of access or retrieval restrictions; that does not establish that they are broken. Section 2.6 also contains unresolved “(?)” citations.

Location in revised manuscript Section 2.6, pp. 23–24; References [1]–[19], pp. 84–86.

Correction required before resubmission Correct Reference [1] metadata against the linked report, identify the original source and provenance of Reference [14] while retaining the repository copy as a reproducibility link, resolve the four “(?)” citations, and finish title-author-year checks for all remaining links in the final compiled PDF. Do not state that all links were verified until these checks pass. Verification source for [1]: <https://static.tti.tamu.edu/tti.tamu.edu/documents/mobility-report-2025.pdf> (title page).

R2.2 Disclosure of preparation assistance

Reviewer comment verbatim The authors must disclose and acknowledge any assistance received in the preparation of this manuscript, including but not limited to editorial, technical, analytical, or writing support. All such contributions must be clearly stated in the Acknowledgments section.

Response Section 6 now acknowledges Eleftheria Fassman for guidance and feedback, Nancy Yuan and Kashif Charania for feedback and edits, Grammarly for grammar and spelling assistance, and Overleaf for manuscript preparation and formatting. It also states that Arnav Jha takes responsibility for the accuracy and integrity of the work.

The acknowledgment should describe actual assistance precisely and should not imply that use of an editor alone verified bibliographic correctness. The author should confirm names, affiliations and roles, and ensure

that editorial, coding, analytical or writing assistance beyond the listed contributions is also disclosed. OpenAI Codex assisted with drafting and checking this response document against the supplied manuscript and project files; any use of that assistance in preparing the revised manuscript should be described accurately as well.

Location in revised manuscript Section 6 Acknowledgments, p. 83.

Correction required before resubmission Confirm the complete assistance history and update the acknowledgment before asserting that disclosure is complete. Suggested addition, if adopted and verified by the author: “OpenAI Codex assisted in preparing the point-by-point response to reviewers and checking its consistency with the manuscript and project files. The author reviewed and verified the resulting response and retained responsibility for its final content.” Do not use this sentence to imply disclosure of any broader assistance history that has not been confirmed.

R2.3 Recent and foundational literature

Reviewer comment verbatim Include enough recent references along with foundational ones. Ensure references directly support your claims. Avoid “padding.” Use credible sources (peer-reviewed journals, reputable books, official reports).

Response The revised Introduction combines foundational traffic and optimization literature with more recent work. Foundational sources include Max-Pressure (2013), SCOOT (1981), the Ising mapping (2014), QAOA (2014), and the original PressLight, MPLight and CoLight publications. Recent entries include a 2024 adaptive traffic-control review, 2025 urbanization and mobility reports, and 2025 quantum transportation literature. Sections 2.4–2.6 connect the adaptive-control citations to the implemented benchmarks.

Reference [11], “Hybrid quantum optimization in the context of minimizing traffic congestion,” is linked to an arXiv preprint and should be identified as such rather than presented as a peer-reviewed article. The literature discussion should distinguish quantum annealing from gate-based QAOA and relate each citation to the specific claim it supports. The revised scope does not require an arbitrary reference count.

Location in revised manuscript Introduction, pp. 5–9; Sections 2.4–2.6, pp. 17–24; Sections 2.8–2.9, pp. 33–41; References, pp. 84–86.

Correction required before resubmission Resolve the unsupported SCOOT citation markers, complete the reference metadata audit in R2.1, and review strong claims such as the “urban arterial standards” assertion on p. 30 for direct source support. Preprint status source: <https://arxiv.org/abs/2504.08275>.

R2.4 Explicit assumptions justification and impact

Reviewer comment verbatim All assumptions (including implicit assumptions) must be explicitly stated and clearly justified. The authors should explain why each assumption is reasonable and discuss its impact on the results and conclusions.

Response The revision makes several modeling choices explicit: regular 2×2 and 3×3 grids and binary NS/EW decisions (Sections 2.2 and 2.9.1); queue and pressure weights (Section 2.3); Seattle count filtering and spatial assignment (Section 2.7); equal left/right shares of the residual turning probability and route limits (Sections 2.7.7–2.7.9); fixed energy weights (Sections 2.8–2.9); frozen transferred RL policies (Section 2.5); and a finite 600-second evaluation horizon with completed-vehicle outcomes (Section 2.10). Section 5 discusses the consequences of regular geometry, small network size, fixed weights, simplified Seattle mapping and simulated quantum execution.

These assumptions serve tractability and controlled comparison, but restrict external validity. Binary phases omit richer intersection movements; symmetric turning and route limits simplify demand; fixed weights can be

poorly matched to low demand or larger grids; and completed-vehicle averages can change when controllers clear different subsets of vehicles. The correlation analysis now directly tests, and limits, the assumption that better instantaneous energy optimization implies better downstream traffic.

Location in revised manuscript Sections 2.2–2.3, pp. 12–17; Section 2.5, pp. 18–22; Section 2.7, pp. 25–32; Sections 2.8–2.10, pp. 33–45; Section 3.5, pp. 63–64; Section 5, pp. 81–83.

Correction required before resubmission A claim that all assumptions are fully justified would be premature. Add an explicit assumption–rationale–impact account covering perfect TraCI state access, fixed-demand conditional inference, route truncation, controller timing and tie-breaking, RL training transfer, parameter calibration frequency, and unmodeled quantum latency. Correct the false determinism assumption in R1.2. Clearly distinguish recommendations from phases actually executed after minimum-green and clearance constraints.

R2.5 Conclusions supported by the evidence

Reviewer comment verbatim Avoid overreaching conclusions that extend beyond what is supported by the data and analysis.

Response Sections 4 and 5 now emphasize conditional performance and acknowledge that no controller dominates all scenarios or metrics. The revision states that QAOA cannot outperform exhaustive enumeration on the shared instantaneous Hamiltonian, that the 3×3 network remains small, that energy–traffic alignment weakens with network size, and that synthetic-noise results are not hardware validation. It also reports unfavorable outcomes and throughput–delay tradeoffs.

The appropriate conclusion is that the tested pressure-based framework and QAOA sampling procedure produced regime-dependent simulated traffic outcomes. The experiments do not establish quantum computational speedup, uniqueness of the sampling benefit to quantum methods, city-scale deployment readiness, or universal superiority to adaptive control.

Location in revised manuscript Section 3.5, pp. 63–64; Section 3.6, pp. 67–68; Sections 4.2.1–4.2.2, pp. 77–80; Section 5, pp. 80–83.

Correction required before resubmission Remove remaining overstatements, including “isolates the quantum contribution” in the abstract on p. 4 and “the use of a quantum annealer still has benefits” on p. 81. Use “simulated gate-based QAOA” consistently for this implementation. Qualify causal explanations of stochasticity as hypotheses unless directly tested, and reconcile all exact-classical claims with R1.2.

R2.6 Tense and third person

Reviewer comment verbatim Verify that you have used past perfect tense and third person throughout the manuscript wherever applicable.

Response The manuscript generally uses third-person descriptions of the study, controllers and author. However, the supplied revision still mixes present and past tense, and a completed manuscript-wide language check cannot yet be certified. Past perfect is appropriate when one past event preceded another; simple past remains appropriate for completed experimental procedures and results, while present tense can describe equations and established facts. Applying past perfect mechanically to every sentence would obscure the chronology.

The final edit should use third person consistently and past perfect where the sequence requires it, for example: “The policies had been trained on the 2×2 network before they were frozen for evaluation.”

Completed results can be stated as “QAOA completed 980 vehicles on average.” This respects the reviewer’s “wherever applicable” qualification while maintaining clear scientific prose.

Location in revised manuscript Manuscript-wide; particularly Section 2.5, p. 19; Section 2.10.3, pp. 44–45; Sections 3–5, pp. 46–83.

Correction required before resubmission Complete the language edit, retain the reviewer’s original spelling “tence” only in the verbatim quotation, remove the duplicate abstract on pp. 1–5, and correct duplicate headings and informal phrasing. Update this response to describe the completed language check only after the manuscript itself has been edited.

Author verification before submission

The responses above can be finalized once the following substantive and editorial discrepancies have been resolved. This section is for the author and should be removed from the submission copy after the checks are complete.

Classical benchmark provenance

The active a7.py scripts in both classical-global directories import and call simulated annealing; the subsequent exhaustive routine measures the gap. Confirm the source revision behind the saved results or regenerate a genuine exact-controller benchmark. This affects the deterministic-baseline description, confidence intervals, effect sizes and quantum-attribution statements.

Circuit and energy convention

Check the cost-unitary derivation against the implementation before claiming mathematical identity. With the stated spin mapping $s = 2x - 1$, computational-basis Z has eigenvalue $-s$. The scoring energy $-h \cdot s - J \cdot s_i \cdot s_j$ therefore corresponds to $+h \cdot Z - J \cdot Z_i \cdot Z_j$. The implemented positive $2\gamma J$ RZ in the CNOT–RZ–CNOT block corresponds to $+J \cdot Z_i \cdot Z_j$ under the stated $\exp(-i\gamma H)$ convention. Reconcile the coupling sign and the omitted J in manuscript Equation 50 with an explicit diagonal-unitary check; any correction affecting generated samples requires reassessing results.

Missing evidence presentation

The PDF contains intended-figure placeholders, including Figures 17–26 used to answer the reviewer. Insert and verify the actual plots. Add the missing bootstrap energy-reduction results and the requested local-heuristic energy comparison.

Statistical reporting

Section 2.10.3 says 50 2×2 runs, whereas the saved analysis data contain 20 runs per α and the tables use 20. Reconcile the provenance and count. Preserve the distinction between bootstrap intervals in Tables 18, 20 and 21 and Student-t intervals in Tables 22 and 23. Explain fixed-demand and frozen-policy uncertainty limits.

References and acknowledgments

Correct the 2025 mobility-report authors; resolve the four missing SCOOT citations; complete link verification; identify the original Seattle source behind the GitHub CSV; confirm all assistance disclosures, including use of Codex in the response preparation.

Manuscript cleanup

Remove the duplicate abstracts and noise headings, correct the controller count, insert figures, narrow residual quantum-advantage language, and perform the requested tense and third-person review. Refresh all response page, section, table and figure references after final compilation.

Project evidence used for the responses

The following paths are relative to the project workspace and are provided for author verification. They are supporting evidence locations, not additional manuscript sections.

grid_2x2/2_classical_global/a7.py and annealer_classical.py; grid_3x3/2_classical_global/a7.py and annealer_classical.py: deployed classical search and separate exhaustive diagnostics.

grid_2x2/3_quantum/annealer_quantum.py; grid_3x3/3_quantum/annealer.py, annealer_hyper.py and annealer_noise.py: circuit construction, parameter search, sample selection and synthetic noise.

quantum_data/quantum_simulation_results.json and
quantum_data_3x3/quantum_3x3_simulation_results.json: 20 saved runs per α condition for both grids.

data_analysis/make_reviewer_results_tables.py; graphs/reviewer_results_tables/: table values, bootstrap procedure and one-sample Hedges' g calculation.

data_analysis/make_reviewer_energy_traffic_figures.py; graphs/reviewer_energy_traffic/: run-level energy-gap analysis, recovery estimates and within-condition correlations.

quantum_hyperparameter_data_3x3/ and quantum_noise_data_3x3/: saved sensitivity results;
seattle_2x2_quantum_data/ and seattle_3x3_quantum_data/: repeated daily-case records.